Supplementary Material

Title: Methods to include environmental impacts in a cost-effectiveness analysis: a case study of PET/CT surveillance imaging after treatment for stage III melanoma

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Supplementary Table 5	Integrated cost-effectiveness analysis sensitivity analysis results exploring the impact of higher values for carbon emissions.

Supplementary Table 1. CHEERS Checklist

Section/topic	Item No	Guidance for reporting	Reported in section
Title			
Title	1	Identify the study as an economic evaluation and specify the interventions being compared.	Title Page
Abstract			
Abstract	2	Provide a structured summary that highlights context, key methods, results, and alternative analyses.	Abstract, pg. 1
Introduction			
Background and objectives	3	Give the context for the study, the study question, and its practical relevance for decision making in policy or practice.	Introduction, pg. 3
Methods			
Health economic analysis plan	4	Indicate whether a health economic analysis plan was developed and where available.	Methods, Overview of methodology, pg. 4
Study population	5	Describe characteristics of the study population (such as age range, demographics, socioeconomic, or clinical characteristics).	Methods, case study, pg. 5
Setting and location	6	Provide relevant contextual information that may influence findings.	Methods, case study, pg. 5
Comparators	7	Describe the interventions or strategies being compared and why chosen.	Methods, decision analytic model, pg. 5
Perspective	8	State the perspective(s) adopted by the study and why chosen.	Methods, decision analytic model, pg. 5
Time horizon	9	State the time horizon for the study and why appropriate.	Methods, decision analytic model, pg. 5
Discount rate	10	Report the discount rate(s) and reason chosen.	Methods, decision analytic model, pg. 5

Section/topic	Item No	Guidance for reporting	Reported in section
Selection of outcomes	11	Describe what outcomes were used as the measure(s) of benefit(s) and harm(s).	Methods, decision analytic model, pg. 5
Measurement of outcomes	12	Describe how outcomes used to capture benefit(s) and harm(s) were measured.	Methods, decision analytic model, pg. 5
Valuation of outcomes	13	Describe the population and methods used to measure and value outcomes.	Methods, decision analytic model, pg. 5
Measurement and valuation of resources and costs	14	Describe how costs were valued.	Methods, decision analytic model, pg. 5
Currency, price date, and conversion	15	Report the dates of the estimated resource quantities and unit costs, plus the currency and year of conversion.	Methods, decision analytic model, pg. 5
Rationale and description of model	16	If modelling is used, describe in detail and why used. Report if the model is publicly available and where it can be accessed.	Methods, decision analytic model, pg. 5; data availability statement, pg. 17
Analytics and assumptions	17	Describe any methods for analysing or statistically transforming data, any extrapolation methods, and approaches for validating any model used.	NA
Characterizing heterogeneity	18	Describe any methods used for estimating how the results of the study vary for subgroups.	NA
Characterizing distributional effects	19	Describe how impacts are distributed across different individuals or adjustments made to reflect priority populations.	NA
Characterizing uncertainty	20	Describe methods to characterise any sources of uncertainty in the analysis.	Methods, Base case cost-effectiveness, pg. 7; Methods, Demonstrating methods, pg. 9
Approach to engagement with patients and others affected by the study	21	Describe any approaches to engage patients or service recipients, the general public, communities, or stakeholders	NA

Section/topic	Item No	Guidance for reporting	Reported in section
		(such as clinicians or payers) in the design of the study.	
Results			
Study parameters	22	Report all analytic inputs (such as values, ranges, references) including uncertainty or distributional assumptions.	Table 1, pg. 27
Summary of main results	23	Report the mean values for the main categories of costs and outcomes of interest and summarise them in the most appropriate overall measure.	Results, pg. 11
Effect of uncertainty	24	Describe how uncertainty about analytic judgments, inputs, or projections affect findings. Report the effect of choice of discount rate and time horizon, if applicable.	Results, pg. 11
Effect of engagement with patients and others affected by the study	25	Report on any difference patient/service recipient, general public, community, or stakeholder involvement made to the approach or findings of the study	NA
Discussion			
Study findings, limitations, generalizability, and current knowledge	26	Report key findings, limitations, ethical or equity considerations not captured, and how these could affect patients, policy, or practice.	Discussion, pg. 13
Other relevant information			
Source of funding	27	Describe how the study was funded and any role of the funder in the identification, design, conduct, and reporting of the analysis	Title page
Conflicts of interest	28	Report authors conflicts of interest according to journal or International Committee of Medical Journal Editors requirements.	Title page

Supplementary Table 2. Discounting of costs, effects, and carbon emissions for primary analyses of each method.

Method	Output	Undiscounted	Discounted[a]
Base case cost-effectiveness analysis	12-monthly cost	$50,632	$42,194
	No routine cost	$48,721	$40,601
	12-monthly effect	0.9265	0.7721
	No routine effect	0.8999	0.7499
Method 1: Report parallel results	12-monthly carbon emissions	1,497 kg	Not discounted
	No routine carbon emissions	1,037 kg	Not discounted
Method 2: ICFCR	-		
Method 3: ICFER	-		
Method 4: Integrated evaluation as a cost	12-monthly healthcare cost	$50,632	$42,194
	No routine healthcare cost	$48,721	$40,601
	12-monthly carbon emissions	$187	$156
	No routine carbon emissions	$129	$108
	12-monthly total cost	$50,819	$42,349
	No routine total cost	$48,850	$40,708

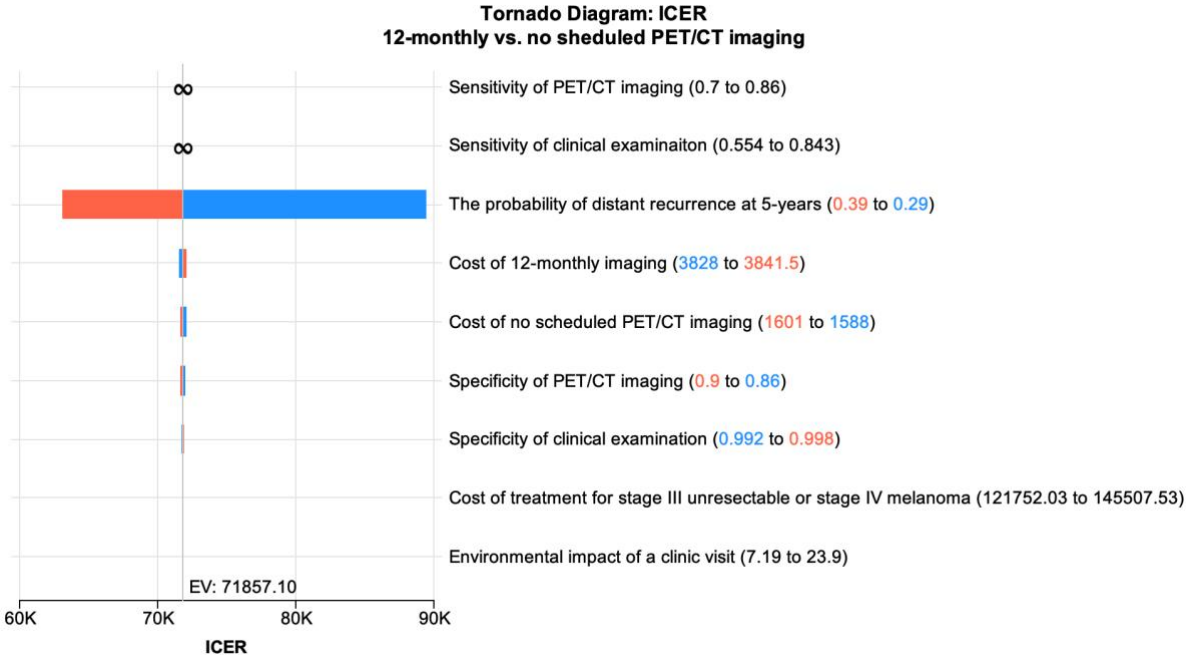
[a] The decision-analytic model structure did not include time or cycles. Discounting was applied to the total five-year expected costs and outcomes manually. A discount rate of 5% per year was applied to costs and health outcomes incurred in years 2-5 of the model.

Supplementary Table 3. Detailed description of assumptions and data inputs for environmental impacts.

Input	Value (CO_2-e)	Assumptions
Environmental impact of 12-monthly scheduled PET/CT imaging	750 kg	Estimate obtained from an existing study by Williams et al. reporting CO_2-e per patient-year (150 kg).[1] This was multiplied by 5, totalling 750 kg CO_2-e.
Environmental impact of no scheduled PET/CT imaging	250 kg	Estimate obtained from an existing study by Williams et al. reporting CO_2-e per patient-year (50 kg).[1] This was multiplied by 5, totalling 250 kg CO_2-e.
Environmental impact of treatment for distant recurrence	2,117.3 kg	The impact of melanoma treatment is not known. In this study we assumed treatment involved two components. First, an estimate of CO_2-e for Nivolumab from a study by Malmberg et al.[2] We assumed 12 months treatment at a dose of 240 mg every two weeks (1892.8 kg). Second, an estimate of CO_2-e for palliative end-of-life care was included. This was based on an overall survival of 71.6% at 12-months with treatment of nivolumab, meaning approximately 28.4% of patients would require palliation at 12-months after distant recurrence.[3] A study by Dokal et al. reports the environmental impact of a 16-bed inpatient hospice as 71.87 kg CO_2-e per bed per day.[4] The average length of stay for palliative care-related overnight hospitalisations in Australia is 11 days.[5] i.e., ((420 / 16 / 365.25) * 11) * 0.284 = 224.5 kg The two components were then added together, totalling 2117.3 kg CO_2-e.
Environmental impact of fine needle aspiration biopsy (FNAB)	0.647 kg	The impact of FNAB is not known. In this study we assumed FNAB had the same impact as a pathology test and one 20-minute ultrasound. An estimate for this was obtained from existing studies by McAlister et al. (0.116 kg for a full blood

		examination and 0.53 kg for a 20-minute ultrasound).[6,7] These two estimates were then added together, totalling 0.647 kg CO_2-e.
Environmental impact of PET/CT	115 kg	The impact of a PET/CT test is not known. In this study we assumed the attributional per minute impact of one PET/CT scan to be the same as the attributional per minute impact of one CT scan. An estimate for this was available from McAlister et al. (1.15 kg).[7] We assumed the PET/CT scan to wake 100 minutes as previously reported by Williams et al. resulting in a total of 115 kg CO_2-e.[1]
Environmental impact of clinic visit	13·9 (7·19-23·9)	Data obtained from an in-press life cycle assessment of a clinic appointment for a full skin examination by a dermatologist. Protocol for this study is available on the Open Science Framework.[8]

Supplementary Figure 1. One-way sensitivity analyses exploring the impact of parameter uncertainty on the ICER.



NOTE: Bars are coloured by parameter range. ∞ refers to variables where the ICER approaches infinity within the specified range. The ICER approaches infinity as the sensitivity of PET/CT imaging approaches 0.726 and the sensitivity of clinical examination approaches 0.82375.

Supplementary Table 4. Sensitivity analyses exploring the impact of higher rates of 5-year distant melanoma recurrence.

		12-monthly	No routine[a]
20% 5-year recurrence	Effect[b]	0.7983	0.7857
	Incremental effect[b]	0.0126	-
	Healthcare cost, AUD	$25,493	$23,772
	Incremental cost, AUD	$1,721	-
	ICER, AUD per effect[b]	$136,587	
35% 5-year recurrence	Effect[b]	0.7721	0.7499
	Incremental effect[b]	0.0222	-
	Healthcare cost, AUD	$42,193	$40,601
	Incremental cost, AUD	$1,593	-
	ICER, AUD per effect[b]	$71,842	-
52% 5-year recurrence	Effect[b]	0.7423	0.7094
	Incremental effect[b]	0.0329	-
	Healthcare cost, AUD	$61,121	$59,673
	Incremental cost, AUD	$1,448	-
	ICER, AUD per effect[b]	$43,975	-
61% 5-year recurrence	Effect[b]	0.7266	0.6879
	Incremental effect[b]	0.0538	-
	Healthcare cost, AUD	$71,141	$69,771
	Incremental cost, AUD	$1,370	-
	ICER, AUD per effect[b]	$25,449	-

AUD: Australian dollars, ICER: incremental cost-effectiveness ratio, ICFCR: incremental carbon footprint cost ratio

[a] Used as comparator for incremental calculation

[b] Proportion of outcomes that are not false negatives

NOTE: Results discounted at 5% per year.

Supplementary Table 5. Integrated cost effectiveness analysis sensitivity analysis results exploring the impact of higher value of carbon emissions.

		12-monthly	No routine[a]
$220.72 AUD per t CO_2-e (NSW Government target consistent approach)	Effect[b]	0.7721	0.7499
	Incremental effect[b]	0.0222	-
	Healthcare cost, AUD	$42,193	$40,601
	Carbon emissions, AUD	$275	$191
	Total costs, AUD[c]	$42,469	$40,792
	Incremental cost, AUD	$1,678	-
	ICER, AUD per effect[b]	75,586	-
$124.75 AUD per t CO_2-e (NSW Government target consistent approach)	Effect[b]	0.7721	0.7499
	Incremental effect[b]	0.0222	-
	Healthcare cost, AUD	$42,193	$40,601
	Carbon emissions, AUD	$156	$108
	Total costs, AUD[c]	$42,349	$40,708
	Incremental cost, AUD	$1,641	-
	ICER, AUD per effect[b]	73,919	-
$86.37 AUD per t CO_2-e (NSW Government target consistent approach)	Effect[b]	0.7721	0.7499
	Incremental effect[b]	0.0222	-
	Healthcare cost, AUD	$42,193	$40,601
	Carbon emissions, AUD	$108	$75
	Total costs, AUD[c]	$42,302	$40,675
	Incremental cost, AUD	$1,626	-
	ICER, AUD per effect[b]	73,243	-
$328.88 AUD per t CO_2-e (Rennert et al. social cost of carbon)	Effect[b]	0.7721	0.7499
	Incremental effect[b]	0.0222	-
	Healthcare cost, AUD	$42,193	$40,601
	Carbon emissions, AUD	$410	$284
	Total costs, AUD[c]	$42,604	$40,885
	Incremental cost, AUD	$1,719	-
	ICER, AUD per effect[b]	77,432	-
$2,055.45 AUD per t CO_2-e (Ducrot et al. mortality-accounting social cost of carbon)	Effect[b]	0.7721	0.7499
	Incremental effect[b]	0.0222	-
	Healthcare cost, AUD	$42,193	$40,601
	Carbon emissions, AUD	$2,565	$1,776
	Total costs, AUD[c]	$44,758	$42,377
	Incremental cost, AUD	$2,382	-
	ICER, AUD per effect[b]	107,297	-

AUD: Australian dollars, ICER: incremental cost-effectiveness ratio, ICFCR: incremental carbon footprint cost ratio

[a] Used as comparator for incremental calculation

[b] Proportion of outcomes that are not false negatives

[c] Total cost is the sum of health care cost and environmental cost

NOTE: Results discounted at 5% per year.

References

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